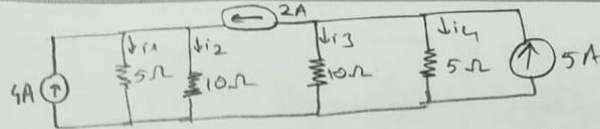


Q1)



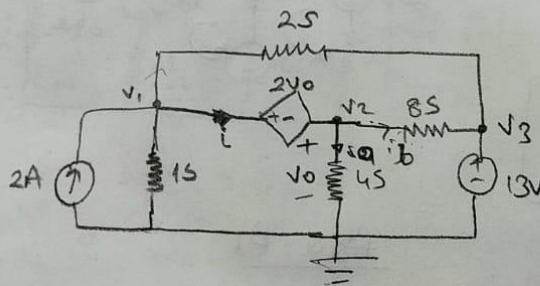
for node 1
KCL
 $i_1 = 4A$

for node 2
KCL
 $i_1 = i_2 + 2$
 $i_2 = 2A$

for node 3
KCL
 $i_2 = i_3 + 2$
 $i_3 = 2A$

for node 4
KCL
 $i_3 = i_4 - 5A$
 $i_4 = 7A$

Q2)



for v_1 $\rightarrow -2 + v_1 + \frac{v_1 - v_3}{2} + i_1 = 0$

for v_2
 $v_2 = v_0$

for i ; $i_a = v_2/4$
 $i_b = \frac{v_2 - v_3}{8}$ $v_3 = 13V$

$-2 + v_1 + \frac{v_1 - 13}{2} + \frac{v_2}{4} + \frac{v_2 - 13}{8} = 0$

$v_1 - v_2 = 2v_0 - v_0 = v_2$

$v_1 = 3v_2$

$-2 + 3v_2 + \frac{3v_2 - 13}{2} + \frac{v_2}{4} + \frac{v_2 - 13}{8} = 0$

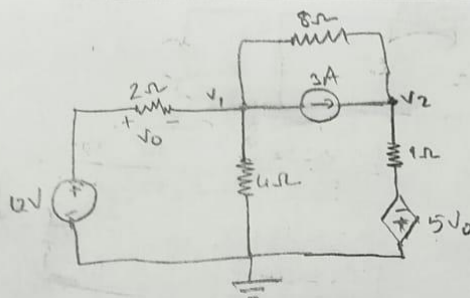
$\rightarrow -2 + \frac{24v_2 + 12v_2 - 52 + 2v_2 + v_2 - 13}{8}$

$39v_2 - 65 = 16$

$39v_2 = 81$

$v_2 = \frac{81}{39}$ $v_1 = \frac{81}{13}$ $v_3 = 13V$

Q3)



$$V_o = V_1 - 12V$$

for V_1) $\frac{V_1 - 12V}{2} + \frac{V_1 - V_2}{8} + \frac{V_1}{4} + 3 = 0$ 1) $4V_1 - 48V + V_1 - V_2 + 2V_1 + 24 = 0$

for V_2) $\frac{V_2 - V_1}{8} - 3 + \frac{V_2 + 5V_o}{1} = 0$ 2) $V_2 - V_1 - 24 + 8V_2 + 40V_o = 0$

$$7V_1 - V_2 = 24V$$

$$V_2 = \frac{840V}{17} - 24V$$

$$V_2 = 25,42V$$

$$V_1 = \frac{120V}{17} = 7,06V$$

Q4) For i_1

$$3i_1 - 2i_2 = -20$$

$$-2i_1 + 4i_2 - 2i_3 = 30$$

$$-2i_2 + 6i_3 = 10$$

$$\begin{bmatrix} 3 & -2 & 0 \\ -2 & 4 & -2 \\ 0 & -2 & 6 \end{bmatrix} \begin{bmatrix} -20 \\ 30 \\ 10 \end{bmatrix}$$

find D

$$\begin{array}{ccc|ccc} 3 & -2 & 0 & 3 & -2 & \\ -2 & 4 & -2 & -2 & 4 & \\ 0 & -2 & 6 & 0 & -2 & \end{array}$$

$$-(0 + 12 + 24) + (72 + 0 + 0)$$

$$-36 + 72 = 36$$

$$3i_1 - 2i_2 = -20$$

$$3(0) - 2i_2 = -20$$

$$i_2 = 10A$$

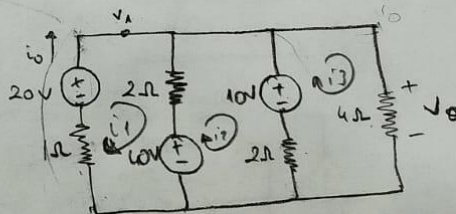
$$-2i_2 + 6i_3 = 10$$

$$i_3 = 5A$$

for loop i_1

$$i_0 = i_1 \text{ so}$$

$$i_0 = 0A$$



find D_1

$$\begin{array}{ccc|ccc} -20 & -2 & 0 & -20 & -2 & \\ 30 & 4 & -2 & 30 & 4 & \\ 10 & -2 & 6 & 10 & -2 & \end{array}$$

$$-(0 - 80 - 360) + (-480 + 40 + 0)$$

$$440 - 440 = 0 \quad D_1 = 0$$

$$i_1 = \frac{D_1}{D} = \frac{0}{36} = 0A$$

for loop i_3

$$V_o = 4\Omega i_3$$

$$V_o = 4\Omega 5A$$

$$V_o = 20V$$

Q5)

for V_1) $-2 + \frac{V_1}{2} + \frac{V_1 - V_3}{4} + \frac{V_1 + V_2}{1} + 3i_o = 0$

for V_2) $\frac{V_2 - V_1}{1} + \frac{V_2 - V_3}{1} + \frac{V_2}{2} = 0$ $i_o = \frac{V_2}{4}$

for V_3) $-4 + \frac{V_3}{2} + \frac{V_3 - V_2}{1} - 3i_o + \frac{V_3 - V_1}{4} = 0$

1... $2V_1 + V_1 - V_3 + 3V_2 + 4V_1 - 4V_2 = 8$

$7V_1 + V_2 - V_3 = 8$

2... $4V_2 - 4V_1 + 4V_2 - 4V_3 + V_2 = 0$

$-4V_1 + 9V_2 - 4V_3 = 0$

3... $16 = 2V_3 + 4V_3 - 4V_2 - 3V_2 + V_3 - V_1$

$-V_1 - 7V_2 + 7V_3 = 16$

for D

$$\begin{bmatrix} 7 & -1 & -1 \\ -4 & 9 & -4 \\ -1 & -7 & 7 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \\ 16 \end{bmatrix}$$

$D = 176$

for D_1

$D_1 = 488$

$V_1 = \frac{D_1}{D} = \frac{488}{176}$

$V_1 = 2.77V$

for D_2

$D_2 = 768$

$V_2 = \frac{D_2}{D} = \frac{768}{176}$

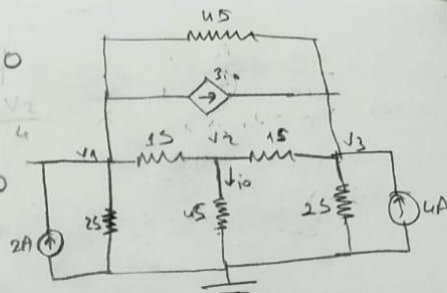
$V_2 = 4.36V$

for D_3

$D_3 = 1240$

$V_3 = \frac{D_3}{D} = \frac{1240}{176}$

$V_3 = 7.05V$



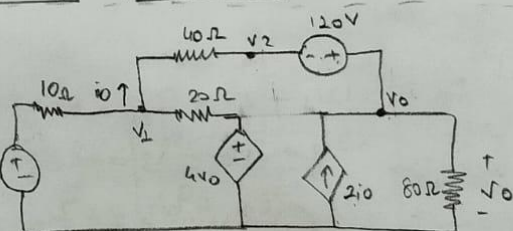
$$\begin{bmatrix} 7 & -1 & -1 \\ -4 & 9 & -4 \\ -1 & -7 & 7 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \\ 16 \end{bmatrix}$$

Q6) $i_o = \frac{V_2 - V_1}{40} = \frac{V_1 - V_0 + 120}{40}$

$V_2 = V_0 - 120$

eq1) $\frac{V_1 - 100}{10} + \frac{V_1 - 4V_0}{20} + \frac{V_1 - V_2}{40} = 0$

eq2) $\frac{V_0 - 120 - V_1}{40} + \frac{V_0}{80} - 2i_o = 0$



1... $4V_1 - 400 + 2V_1 - 8V_0 + V_1 - V_2 = 0$ $V_2 = V_0 - 120$ $-8V_0 + 7V_1 - V_0 + 120 = 400$

$-8V_0 + 7V_1 = 280$

2... $2V_0 - 240 - 2V_1 + V_0 - 160i_o = 0$ $V_2 = V_0 - 120$ $3V_0 + 2V_1 - 4(V_0 - 120) = 240$

$-V_0 + 2V_1 = -240$

$3V_0 - 2V_1 = 160i_o = 240$

$3V_0 - 2V_1 - 4V_2 + 4V_1 = 240$

$3V_0 + 2V_1 - 4V_2 = 240$

$V_0 = -9 \cdot 280 + 7 \cdot (-240)$

$V_0 = -4200V$

$V_1 = -280 - 480$

$V_1 = -760V$

$i_o = \frac{V_2 - V_1}{40} = \frac{V_0 - 120 - V_1}{40} = \frac{-4200V - 120 + 760}{40}$

$i_o = -89A$

$$\begin{bmatrix} -9 & 7 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 280 \\ -240 \end{bmatrix}$$

Q7)

$$1 + 2V_0 = \frac{V_1}{4} + \frac{V_2}{1} + \frac{V_1 - V_3}{1} \quad (1)$$

and $V_0 = V_1 - V_3$

$$u = -3V_1 + 4V_2 + 4V_3$$

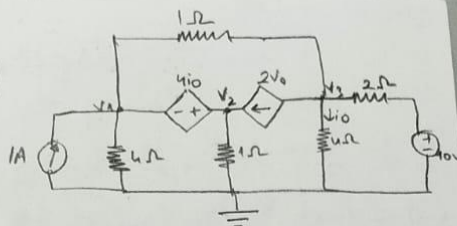
$$\frac{2V_0 + V_3}{4} = \frac{V_1 - V_3 + 10 - V_3}{2}$$

$$20 = 4V_1 + 0V_2 - V_3 \quad (3)$$

and $V_2 = V_1 + u_{10}$ and $i_0 = \frac{V_3}{4}$

$$V_2 = V_1 + V_3 \quad (4)$$

$$V_1 = 4.97V \quad V_2 = 4.85V \quad V_3 = -0.12V$$



Q8)

for ii) $12i_1 - 2i_2 + i_3 = 6$

$$-2i_1 + 7i_2 - i_3 = -8$$

$$0i_1 - i_2 + 6i_3 = +2$$

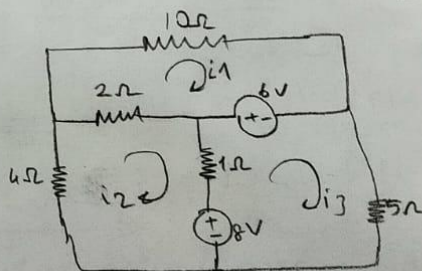
$$\begin{bmatrix} 12 & -2 & 0 \\ -2 & 7 & -1 \\ 0 & -1 & 6 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 2 \end{bmatrix}$$

$$\begin{array}{ccccccc} 12 & -2 & 0 & 12 & -2 & & \\ -2 & 7 & -1 & -2 & 7 & & \\ 0 & -1 & 6 & 0 & -1 & & \\ 0 & 12 & -2 & 50 & -2 & & \\ -12 & 50 & -2 & & & & \end{array} \quad D = 492$$

for D2

$$12i_1 - 2i_2 = 6$$

$$i_2 = 6i_1 - 3 = -1.12A$$



for D1

$$\begin{array}{ccccccc} 6 & -2 & 0 & +6 & -2 & & \\ -8 & 7 & -1 & -8 & 7 & & \\ 2 & -1 & 6 & 2 & -1 & & \\ -(0+6+96) & + & (252+4+0) & & & & \\ -102 & + & 256 & = & 154 & & \end{array} \quad D_1 = 154$$

for D3

$$-i_2 + 6i_3 = 2$$

$$i_3 = \frac{2 + i_2}{6} = 0.147A$$

Q9) $40i_1 - 10i_2 + 0i_3 = -120$
 $-10i_1 + 50i_2 - 10i_3 = 0$
 $0i_1 - 10i_2 + 40i_3 = 120$

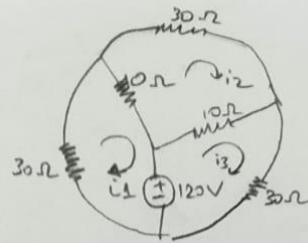
$$\begin{bmatrix} 40 & -10 & 0 \\ -10 & 50 & -10 \\ 0 & -10 & 40 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} -120 \\ 0 \\ 120 \end{bmatrix}$$

for D

$$\begin{vmatrix} 40 & -10 & 0 \\ -10 & 50 & -10 \\ 0 & -10 & 40 \end{vmatrix} = 40(2000 - 4000) + (80000 + 0 + 0) = 72000$$

$$-(0 + 2000 + 4000) + (80000 + 0 + 0) = 72000$$

$$\begin{aligned} i_1 &= -3 \\ i_2 &= 0 \\ i_3 &= 3 \end{aligned} \quad \begin{aligned} 40i_1 - 10i_2 &= -12 \\ -12 - 10i_2 &= -12 \end{aligned}$$



for D1

$$\begin{vmatrix} -120 & -10 & 0 & -120 & -10 \\ 0 & 50 & -10 & 0 & 50 \\ 120 & -10 & 40 & 120 & -10 \end{vmatrix}$$

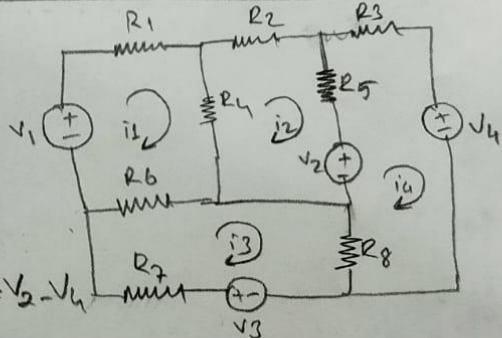
$$-(0 - 12000 + 0) + (240000 + 12000)$$

$$D_1 = -216000$$

$$i_1 = \frac{D_1}{D} = \frac{-216000}{72000} = -3$$

Q10)

$$\begin{aligned} i_1(R_1 + R_4 + R_6) - i_2 R_4 - i_3 R_6 &= V_1 \\ -i_1 R_4 + i_2(R_2 + R_4 + R_5) - i_4 R_5 &= 0 \\ -i_1 R_6 + i_3(R_6 + R_7 + R_8) - i_4 R_8 &= V_3 \\ -i_2(R_5) - i_3 R_8 + i_4(R_3 + R_4 + R_5 + R_8) &= V_2 - V_4 \end{aligned}$$



Q11)

$$\begin{aligned} i_1 \cdot 9 - 3i_2 - 4i_3 &= 6 \\ -3i_1 + 8i_2 &= 4 \\ -4i_1 + 6i_3 - i_4 &= 2 \\ -i_3 + 2i_4 &= -3 \end{aligned}$$

